

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## TAGGATE – Smart Community & Property Management Platform

**Version:** 1.0

**Prepared For:** TAG

**Document Type:** Software Requirements Specification

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## 1. Introduction

### 1.1 Purpose

This document defines the functional and non-functional requirements for the TAGGATE platform.

TAGGATE is a centralized community and property management platform designed for residential societies, villa communities, gated townships, and future smart residential developments.

The platform enables community management, security operations, tenant management, resident services, property monitoring, and future smart home integrations through a unified system.

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### 1.2 Scope

TAGGATE shall provide functionality for:

- User Management
- Property Management
- Resident Management

- Tenant Management
  - Security Management
  - Visitor Management
  - Complaint Management
  - Community Communication
  - Home Services Marketplace
  - Grocery Marketplace
  - NRI Property Management
  - Utility Monitoring
  - Conversational Community Assistant
  - Smart Home Integrations (Future Phase)
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## 2. System Users

### Administrator

Responsible for overall community management.

#### Permissions

- Manage residents
  - Manage properties
  - Manage tenants
  - Manage complaints
  - Manage vendors
  - Publish notices
  - Access reports
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### Resident

#### Permissions

- Raise complaints
- Book services
- Order groceries
- Approve visitors
- View notices

- View utility information
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## Property Owner

### Permissions

- Manage owned properties
  - Add tenants
  - View maintenance history
  - Track property information
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## NRI Owner

### Permissions

- Monitor property remotely
  - Track occupancy
  - View maintenance updates
  - Receive property reports
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## Tenant

### Permissions

- Access community services
  - Raise complaints
  - Manage visitors
  - View notices
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## Security Personnel

### Permissions

- Register visitors

- Verify entries
  - Manage domestic staff
  - Record incidents
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## **Service Provider**

### **Permissions**

- View bookings
  - Accept jobs
  - Update service status
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## **3. Functional Requirements**

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## **Module 1: Authentication & User Management**

### **FR-001**

The system shall support login through:

- Mobile Number + OTP
- Email + Password

### **FR-002**

The system shall provide role-based access control.

### **FR-003**

Users shall be able to update profile information.

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## Module 2: Resident Management

### FR-004

The system shall maintain resident records.

### FR-005

Residents shall be associated with properties.

### FR-006

Administrators shall activate/deactivate resident accounts.

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## Module 3: Property Management

### FR-007

The system shall maintain property records.

### FR-008

Property information shall include:

- Property Number
- Property Type
- Owner Information
- Occupancy Status

### FR-009

One owner may own multiple properties.

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## Module 4: Tenant Management

### FR-010

Owners shall add tenant details.

### **FR-011**

The system shall support tenant document uploads.

Examples:

- Aadhaar
- PAN
- Passport
- Rental Agreement

### **FR-012**

Administrators shall approve tenant onboarding.

### **FR-013**

The system shall track:

- Move-In Date
  - Move-Out Date
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## **Module 5: Security Management**

### **FR-014**

Residents shall approve visitors.

### **FR-015**

Security personnel shall register visitors.

### **FR-016**

The system shall maintain visitor logs.

### **FR-017**

The system shall manage:

- Domestic Staff
- Drivers
- Maids
- Service Personnel

#### **FR-018**

Vehicle records shall be maintained.

#### **FR-019**

Security incidents shall be recorded.

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## **Module 6: Complaint Management**

#### **FR-020**

Residents shall create complaints.

#### **FR-021**

Complaint information shall include:

- Category
- Description
- Attachments
- Status

#### **FR-022**

Administrators shall assign complaints.

#### **FR-023**

Residents shall track complaint status.

#### **FR-024**

Complaint history shall be maintained.

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## **Module 7: Community Communication**

### **FR-025**

Administrators shall publish notices.

### **FR-026**

Residents shall receive notifications.

### **FR-027**

Community announcements shall be archived.

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## **Module 8: NRI Property Management**

### **FR-028**

NRI owners shall view:

- Occupancy Information
- Property Status
- Maintenance Activities
- Property Updates

### **FR-029**

Inspection reports shall be maintained.

### **FR-030**

Maintenance history shall be available.

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## **Module 9: Home Services Marketplace**

### **FR-031**



Residents shall browse service categories.

Examples:

- Plumbing
- Electrical
- Housekeeping
- AC Service
- Pest Control

### **FR-032**

Residents shall create service bookings.

### **FR-033**

Service providers shall accept bookings.

### **FR-034**

Residents shall track service requests.

### **FR-035**

Residents shall rate completed services.

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## **Module 10: Grocery Marketplace**

### **FR-036**

Residents shall browse products.

### **FR-037**

Residents shall add products to cart.

### **FR-038**

Residents shall place orders.

### **FR-039**

Residents shall track order status.

#### **FR-040**

Order history shall be maintained.

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## **Module 11: Utility Monitoring**

#### **FR-041**

Residents shall view:

- Electricity Usage
- Water Usage
- Maintenance Charges

#### **FR-042**

The system shall display:

- Daily Usage
- Weekly Usage
- Monthly Usage

#### **FR-043**

Residents shall receive payment reminders.

#### **FR-044**

Residents shall receive usage alerts.

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## **Module 12: Conversational Community Assistant**

### **Overview**

The platform shall provide a conversational interface allowing residents to interact with community services through natural language.

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#### **FR-045**

Residents shall access the assistant from web and mobile applications.

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#### **FR-046**

Residents shall raise complaints through chat.

Example:

Water leakage in kitchen.

System Action:

- Create complaint
  - Assign category
  - Generate ticket
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#### **FR-047**

Residents shall book services through chat.

Example:

Book an electrician tomorrow.

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#### **FR-048**

Residents shall order groceries through chat.

Example:

Order 2 litres milk and 1 kg tomatoes.

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### **FR-049**

Residents shall check utility information through chat.

Example:

Show electricity usage this week.

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### **FR-050**

Residents shall check maintenance dues through chat.

Example:

How much maintenance is pending?

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### **FR-051**

Residents shall request community information through chat.

Examples:

- Clubhouse timings
  - Visitor policies
  - Tenant registration process
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### **FR-052**

The assistant shall automatically categorize complaints.

Examples:

- Plumbing
  - Electrical
  - Appliance
  - Security
  - Housekeeping
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### **FR-053**

The assistant shall automatically create tickets from complaints.

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## **Module 13: Smart Appliance Monitoring (Future Phase)**

### **FR-054**

The platform shall integrate with supported smart devices.

Examples:

- Smart Meters
  - Air Conditioners
  - Water Pumps
  - Water Purifiers
  - Solar Systems
  - Water Tank Sensors
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### **FR-055**

The platform shall monitor appliance health.

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### **FR-056**

The platform shall detect abnormal appliance behavior.

Examples:

- Excessive power consumption
  - Continuous motor operation
  - Sensor abnormalities
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### **FR-057**

The platform shall notify residents of detected issues.

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#### **FR-058**

The platform shall generate maintenance tickets automatically when appliance issues are detected.

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#### **FR-059**

The platform shall notify maintenance teams of generated tickets.

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## **Module 14: Reports & Dashboard**

#### **FR-060**

Administrators shall access dashboards.

#### **FR-061**

Reports shall be available for:

- Residents
- Properties
- Tenants
- Complaints
- Visitors
- Services
- Utility Usage

#### **FR-062**

Reports shall support export functionality.

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## **4. Non-Functional Requirements**

## **Performance**

### **NFR-001**

System response time shall be less than 3 seconds for normal operations.

### **NFR-002**

The system shall support multiple communities simultaneously.

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## **Security**

### **NFR-003**

Role-based access control shall be implemented.

### **NFR-004**

Passwords and sensitive information shall be encrypted.

### **NFR-005**

All communication shall use secure protocols.

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## **Scalability**

### **NFR-006**

The platform shall support onboarding multiple societies and villa projects.

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## **Availability**

### **NFR-007**

System availability shall be maintained at 99.5% or higher.

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## Usability

### NFR-008

The platform shall be available through:

- Web Application
- Android Application
- iOS Application

### NFR-009

The interface shall remain simple and user-friendly.

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## 5. Future Integrations

The platform should support integration with:

- Smart Home Devices
  - IoT Sensors
  - Utility Monitoring Systems
  - CCTV Systems
  - Access Control Systems
  - WhatsApp Notifications
  - Payment Gateways
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## 6. Implementation Phases

### Phase 1

- Resident Management
- Property Management
- Tenant Management
- Security Management



- Complaint Management
- Community Communication

## **Phase 2**

- NRI Property Management
- Home Services Marketplace
- Grocery Marketplace
- Utility Monitoring

## **Phase 3**

- Conversational Community Assistant
- Smart Appliance Monitoring
- Automatic Ticket Generation
- Smart Home Integrations